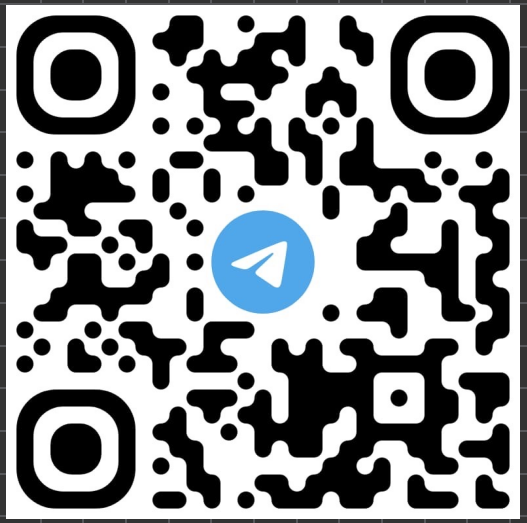
Telegram



Discord



Async JS

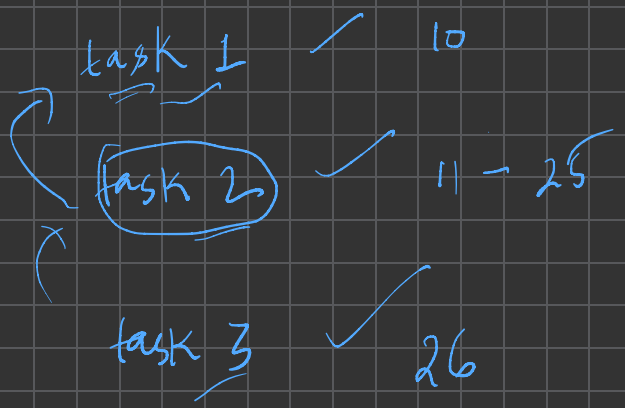
Sync

Syncronization

Async

Asyncronization

Sync



Async task

aisa kaam jo kuch time lega execute
hone mai & uske liye JS wait
nhi karti.

Burger shop

Sync

→ order

→ wait

→ recieve

Async

→ order

→ kuch or kam karlo

→ recieve

JS is single threaded language



Thread

↳ sequence of instructions a computer executes

```
console.log("Task 1");

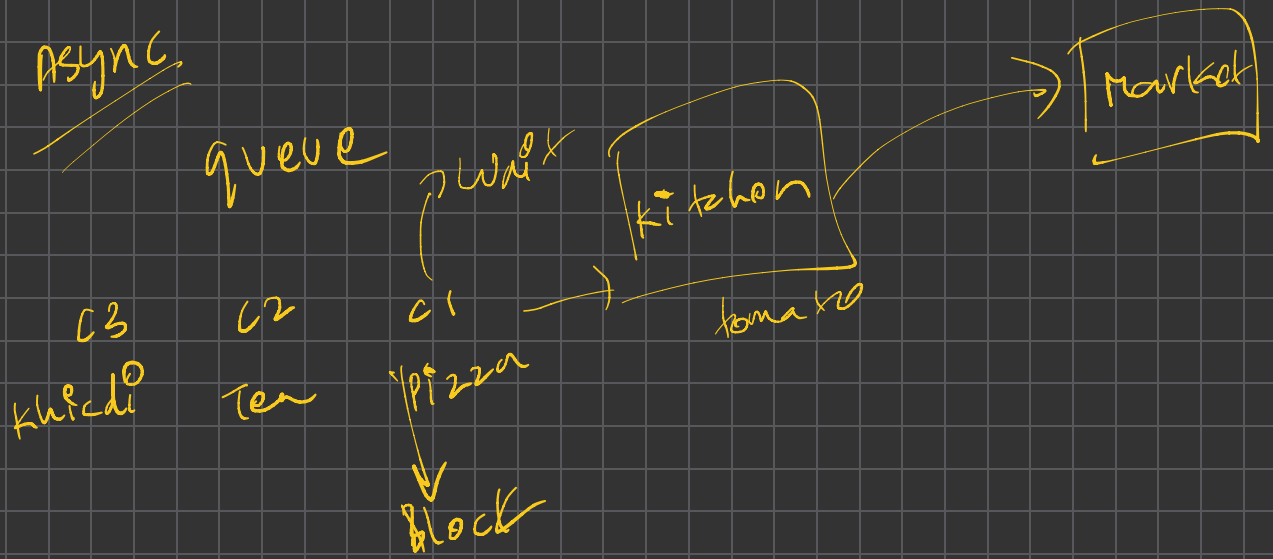
console.log("Task 2");

for(let i = 0 ; i < 10000000000 ; i++){

}

console.log("Task 3");
```

unneccessary main thread ko block kar raha hai


async
queue
c3
khichdi
c2
tea
c1
pizza
block
wait
kitchen
tomato
market

current Unix timestamp in millisecond

```
let startTime = Date.now()

while(Date.now() - startTime < 10000){

}
```

1s = 1000ms

1 Jan 1970

1000

StartTime = 1000 ms

True

0 < 10000

while(1000 - 1000 < 10000){

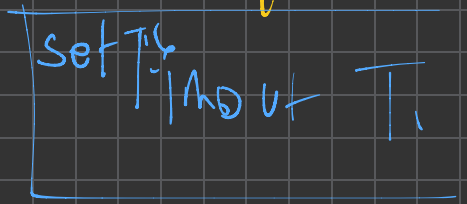
3

while(2000 - 1000 < 10000)
(1000 <= 10000)

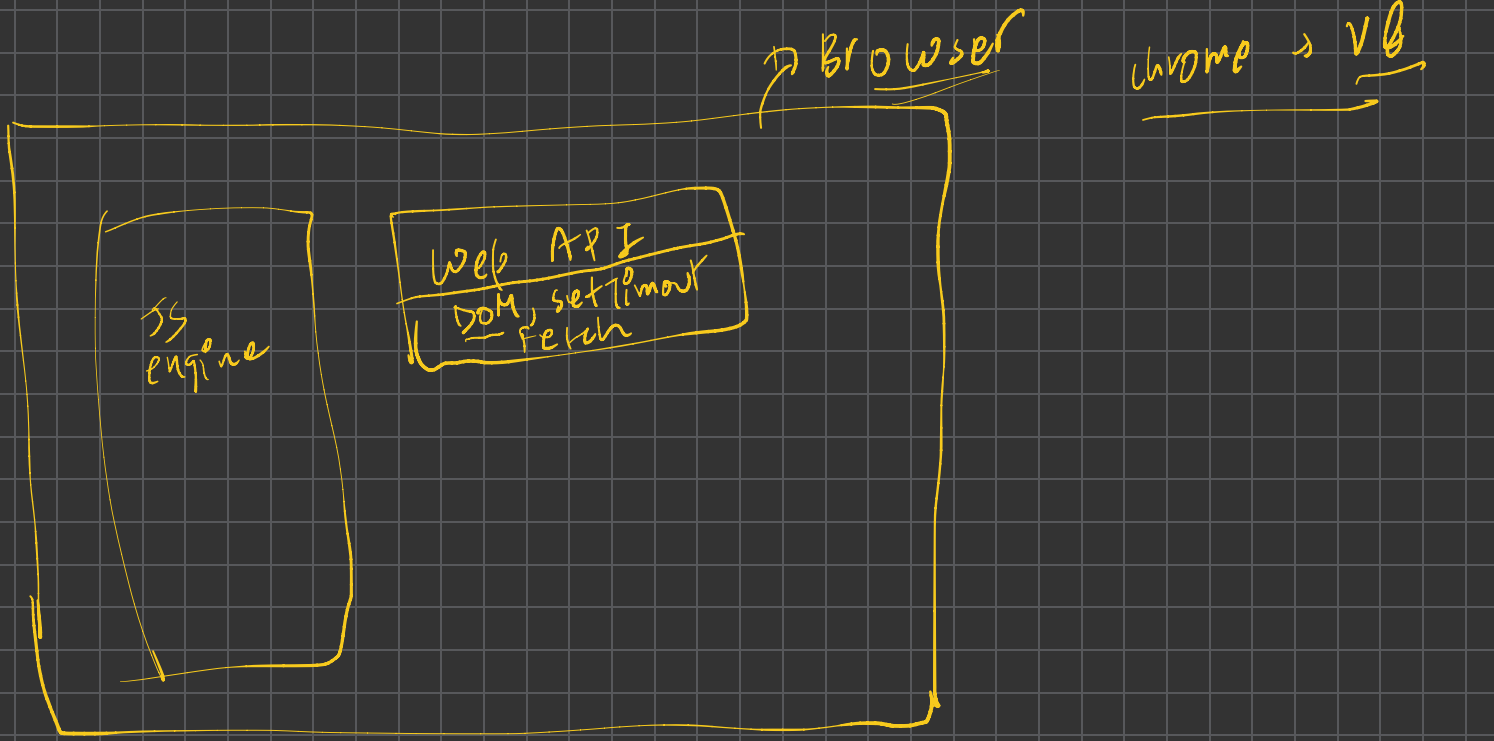
11000 - 1000 < 10000
10000 < 10000
false

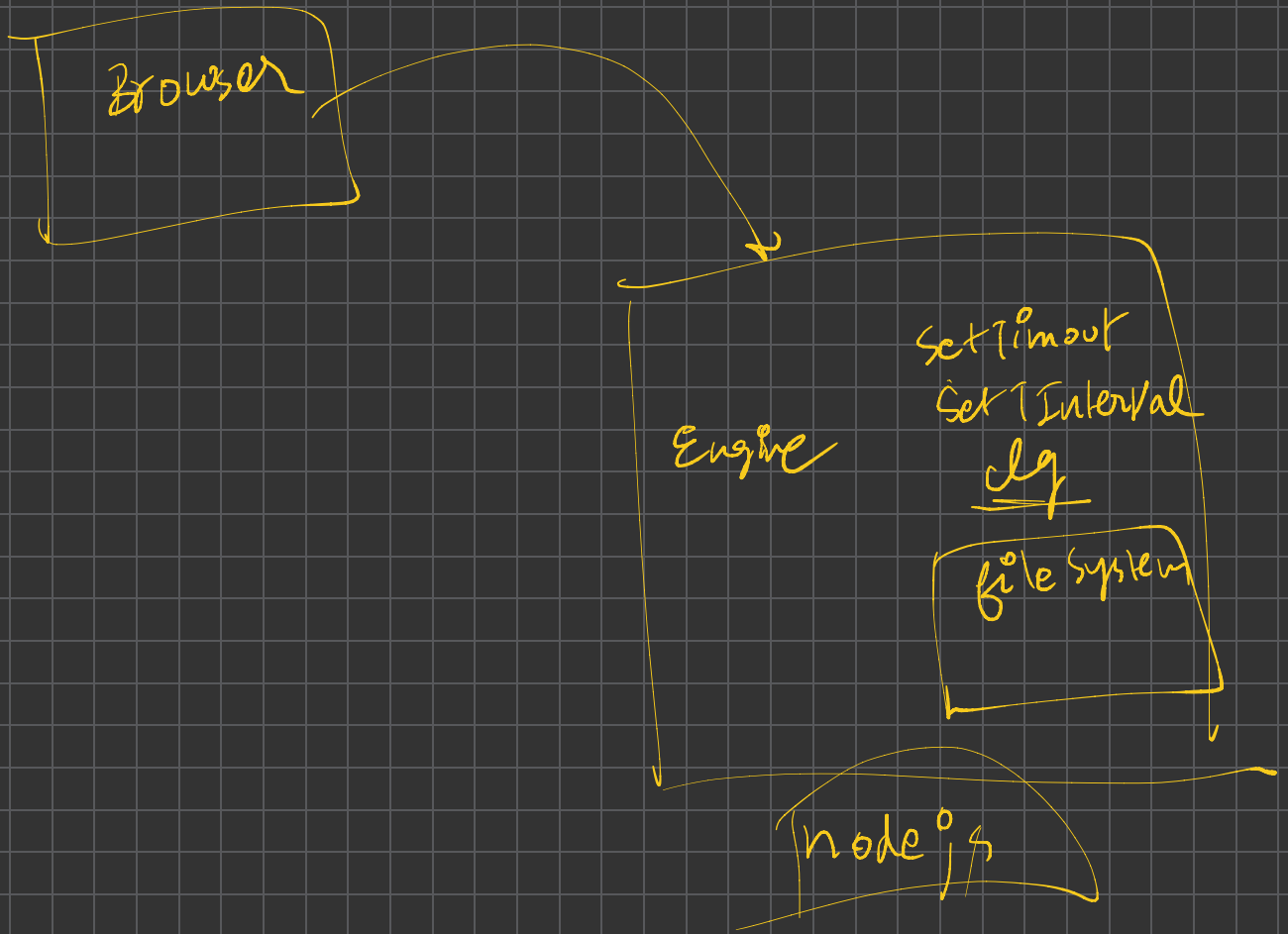
100 files

file 1 → blocking code




Web apis
DOM
console.log
Is ka part nhi hai


Browser
chrome → V8
JS engine
Web API
DOM, setTimout
Fetch


Browser
Engine
setTimout
setTInterval
dg
file system
node js

Web API

- setTimeout
- setInterval

Event Loop

setTimeout(callbackfn , timer)

10,000 → 10sec

↓

in ms

timer → means it takes minimum x time to execute cb function

(4s, 10s)

```
console.log("task 1");

setTimeout(function cb() {
    console.log("task 2");
}, 0)

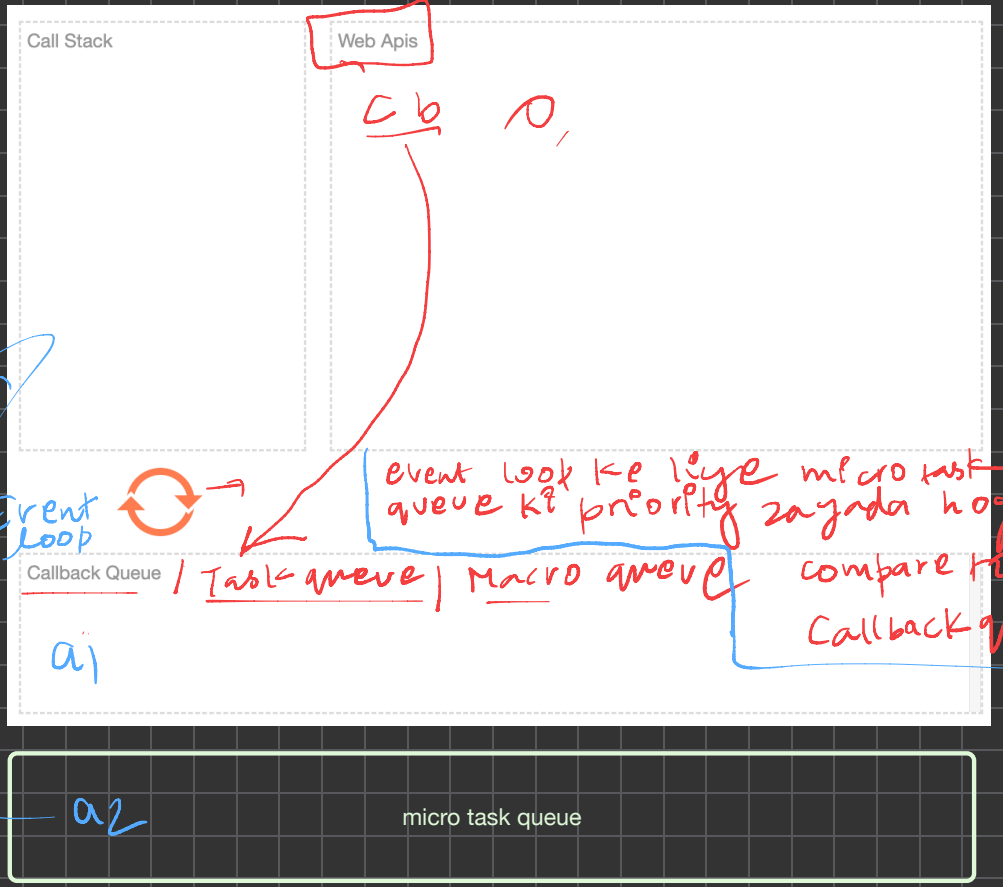
console.log("task 3");
```

output

task 1

task 3

task 2



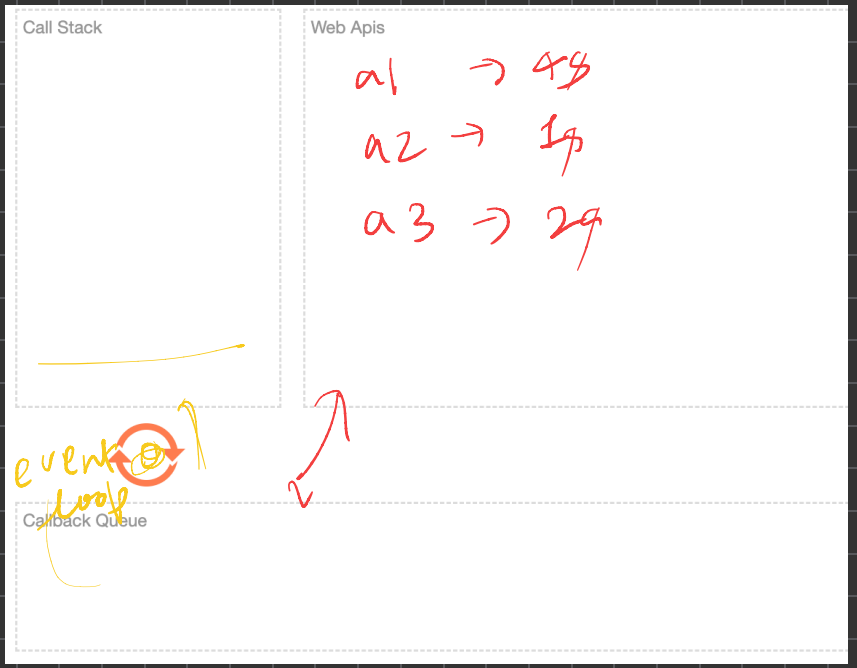
```
console.log("task 1");

setTimeout(()=>{
    console.log("task 2");
}, 4000)

setTimeout(()=>{
    console.log("task 5");
}, 1000)

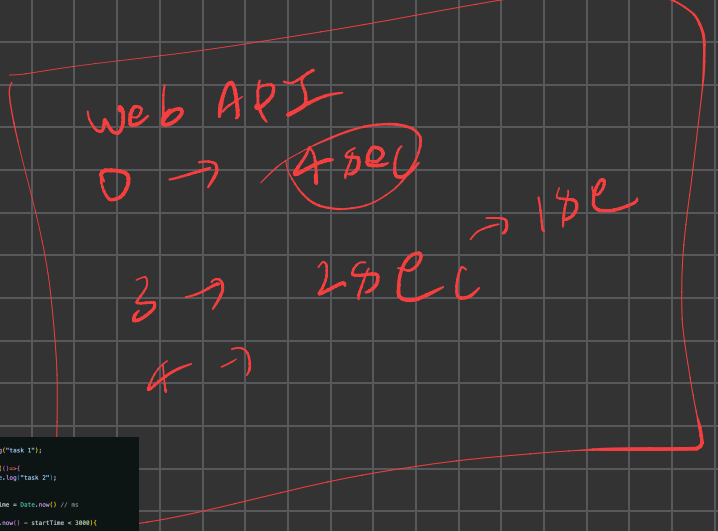
setTimeout(()=>{
    console.log("task 4");
}, 2000)

console.log("task 3");
```



output

1
3
5
4
2




```
console.log("task 1");

setTimeout(()=>{
    console.log("task 2");
}, 4000)

let startTime = Date.now() // ms

while(Date.now() - startTime < 3000){

}

setTimeout(()=>{
    console.log("task 5");
}, 1000)

setTimeout(()=>{
    console.log("task 4");
}, 2000)

console.log("task 3");
```

```
console.log("task 1");

setTimeout(()=>{
    console.log("task 2");
}, 4000)

let startTime = Date.now() // ms

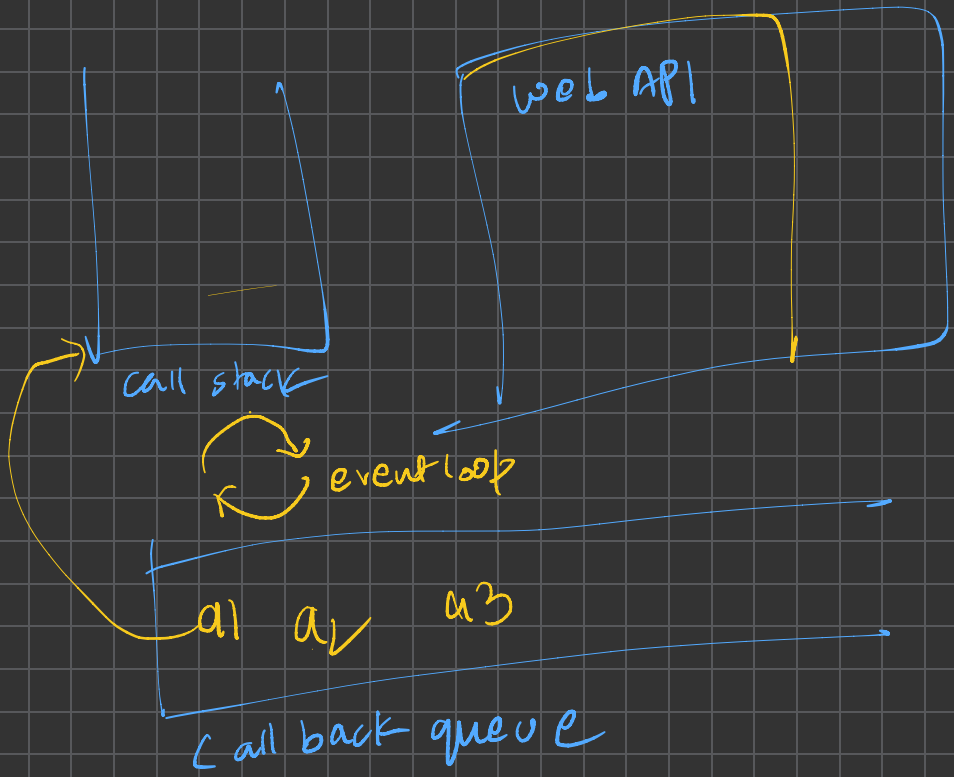
while(Date.now() - startTime < 3000){

}

setTimeout(()=>{
    console.log("task 5");
}, 1000)

setTimeout(()=>{
    console.log("task 4");
}, 2000)

console.log("task 3");
```



output → 1
3
2
5

setInterval (callback fn , Timer)

Timer → ms

callback function again & again Timer ke base per execute hota rahega

0-9

a-f

0 1 2 3 4 5 6 7 8 9 a b c d e f

Nishant

clr = ""

clr = clr "

"N" + "O" + "S" + "H"

(Nish)

let name = "" nish

① → name = "" + "n" → "n"

② name = "n" + "i" → "ni"

③ name = "ni" + "s" → "nis"